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Showing weak stationarity

b) Show $E[X_t]$ has mean not dep. on t

$$E[X_t] = E[Z_0\cos(ct)] + E[Z_1\sin(ct)]$$
$$= \cos(ct)E[Z_0] + \sin(ct)E[Z_1]$$
$$= 0 + 0$$
$$= 0$$

$\therefore$ constant $\forall t$ ✓

Showing $\gamma(h)$ does not dep. on t

$$\gamma(h) = \text{Cov}(X_t, X_{t+h})$$
$$= E[X_t X_{t+h}] \quad (\text{since } E[X_t] = 0, \text{ same reason as 2a})$$
$$= E[[Z_0\cos(ct) + Z_1\sin(ct)][Z_0\cos(c(t+h)) + Z_1\sin(c(t+h))]]$$
$$= E[Z_0^2\cos(ct)\cos(c(t+h)) + Z_0Z_1\cos(ct)\sin(c(t+h))$$
$$+ Z_0Z_1\sin(ct)\cos(c(t+h)) + Z_1^2\sin(ct)\sin(c(t+h))]$$

(simplifying the Z_0Z_1 terms $= 0$ b/c $E[Z_0Z_1] = 0$ by indep. and def'n, and $E[Z_0^2] = E[Z_1^2] = \sigma^2$ by def'n)

$$= \sigma^2\cos(ct)\cos(c(t+h)) + \sigma^2\sin(ct)\sin(c(t+h))$$
$$= \sigma^2[\cos(ct - (c(t+h)))]$$
$$= \sigma^2\cos(ct - ct + ch)$$
$$= \sigma^2\cos(ch).$$

$\therefore \gamma(h)$ does not dep. on t $\forall t$

Thus, we have shown X_t is weakly stationary.

■

Calculating ACF

$\gamma(h) = \sigma^2\cos(ch)$ as above

$$\gamma(0) = \sigma^2\cos(c0)$$
$$= \sigma^2$$

$$\Rightarrow \rho(h) = \frac{\sigma^2\cos(ch)}{\sigma^2}$$

$\therefore \rho(h) = \cos(ch)$ is the ACF of X_t